

Jaehyeon Moon

Yonsei University

Combined M.S. & Ph.D. student

Email

jhmn@yonsei.ac.kr

Website

<https://jaehyeonmoon.github.io/>

Location

50, Yonsei-ro, Seodaemun-gu, Seoul, Korea, 03722

Research Interests

- Model quantization
- KV cache compression

Education

Mar. 2020 – Present **Yonsei University**
School of Electrical and Electronic Engineering
Seoul, Korea
Advisor: Bumsub Ham
Combined M.S. & Ph.D. student

Mar. 2016 – Feb. 2020 **Yonsei University**
School of Electrical and Electronic Engineering
Seoul, Korea
B.S. in Electrical and Electronic Engineering
Magna Cum Laude (GPA: 4.06 / 4.30, Top 3%)

Publications

- **Jaehyeon Moon** and Bumsub Ham, "Shift-and-Sum Quantization for Visual Autoregressive Models", ICLR, 2026.
- Dohyung Kim*, **Jaehyeon Moon***, Junghyup Lee, Geon Lee, Jeimin Jeon, and Bumsub Ham, "Token-Based Dynamic Bit-Width Assignment for ViT Quantization", Pattern Recognition, 2026.
*Equal contribution.
- **Jaehyeon Moon**, Dohyung Kim, Junyong Cheon, and Bumsub Ham, "Instance-Aware Group Quantization for Vision Transformers", CVPR, 2024.
- Seunghoon Lee*, Jeongwoo Choi*, Byungwan Son, **Jaehyeon Moon**, Jeimin Jeon, and Bumsub Ham, "AccuQuant: Simulating Multiple Denoising steps for Quantizing Diffusion Models", NeurIPS, 2025. *Equal contribution.
- Dohyung Kim, Junghyup Lee, Jeimin Jeon, **Jaehyeon Moon**, and Bumsub Ham, "Toward INT4 Fixed-Point Training via Exploring Quantization Error for Gradients", ECCV, 2024.

Experiences

Aug. 2023 – Present **TIPS project**
Research engineer, Articon

Developed quantization and pruning algorithms to deploy deep neural networks on compute-in-memory (CIM) chips under the TIPS startup R&D program

Patents

- Bumsub Ham, **Jaehyeon Moon**, Dohyung Kim, and Junyong Cheon, "Quantization Apparatus and Method for Artificial Neural Networks and Image Processing Device Having the Same", US Patent No. 12,684,127 (issued Jul. 2026)
- Bumsub Ham and **Jaehyeon Moon**, "Quantization Method and Apparatus for Visual Autoregressive Model", US-Application No. 19/548,809
- Bumsub Ham and **Jaehyeon Moon**, "Quantization Method and Apparatus for Visual Autoregressive Model", KR-Application No. 10-2026-0032167
- Bumsub Ham, Dohyung Kim, Junghyup Lee, **Jaehyeon Moon**, Geon Lee, and Jeimin Jeon, "Token-Based Quantization Apparatus and Method for Vision Transformers", KR-Application No. 10-2024-0137421
- Bumsub Ham, Dohyung Kim, Junghyup Lee, Jeimin Jeon, and **Jaehyeon Moon**, "Quantization Apparatus and Method for Artificial Neural Networks", KR-Application No. 10-2023-0116857
- Bumsub Ham, **Jaehyeon Moon**, Dohyung Kim, and Junyong Cheon, "Quantization Apparatus and Method for Artificial Neural Networks and Image Processing Device Having the Same", KR-Application No. 10-2023-0072090

Awards & Scholarships

- National Science & Technology Scholarships, Ministry of Science and ICT, Republic of Korea (Sep. 2018 – Feb. 2020)